COMP7024 Assignment

Vijay Lakshmi (ID - 22193558)

2026-03-02

By including this statement, we the authors of this work, verify that:

- We hold a copy of this assignment that we can produce if the original is lost or damaged.
- We hereby certify that no part of this assignment/product has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.
- No part of this assignment/product has been written/produced for us by another person except where such collaboration has been authorised by the subject lecturer/ tutor concerned.
- We are aware that this work may be reproduced and submitted to plagiarism detection software programs for the purpose of detecting possible plagiarism (which may retain a copy on its database for future plagiarism checking).
- We hereby certify that we have read and understand what the School of Computing, Engineering and Mathematics defines as minor and substantial breaches of misconduct as outlined in the learning guide for this unit.

Loading the data

```r
# Loading orders data

orders <- read.csv("project_orders.csv")
```

```r
# Loading product data

product <- read.csv("project_product_info.csv")
```

Data testing

Checking the dimensions: It should exactly match the numbers of rows in csv file, which shows that whole data has been completely loaded or not.

```r
dim(orders)
```

```
## [1] 541909      7
```

```r
dim(product)
```

```
## [1] 4070    2
```

- The dimensions are exactly equal to the number of rows in the csv file, hence data has been completely loaded.

Question 1: Data Preparation & Product Revenue Analysis

1.1. Data cleaning - Removing cancelled Invoices

- For revenue calculation accuracy it is important to remove cancelled invoices, cancelled invoices are identified by invoice number starting with letter "C".

Code

- Cleaning data by removing cancelled Invoices

```
# Data Cleaning

orders_clean <- orders[substr(orders$InvoiceNo,1,1) !="C",]
```

- Comparing the dimensions after removing cancelled invoices

```
dim(orders)          #original orders
```

```
## [1] 541909      7
```

```
dim(orders_clean)   # cleaned orders
```

```
## [1] 532621      7
```

Interpretation

- The original data set contain 541909 observations. After removing cancelled invoices we are left with 532621 observations. Hence, 9288 invoices are not included in the final revenue calculation.

1.2. Total revenue for each product

- Revenue = Quantity X Unit Price

```
orders_clean$Revenue <- orders_clean$Quantity * orders_clean$UnitPrice

# using head function to check if revenue column appearing or not

head(orders_clean)
```

```
##   InvoiceNo StockCode Quantity    InvoiceDate UnitPrice CustomerID
## 1    536365    85123A        6 1/12/2010 8:26      2.55      17850
## 2    536365     71053        6 1/12/2010 8:26      3.39      17850
## 3    536365    84406B        8 1/12/2010 8:26      2.75      17850
## 4    536365    84029G        6 1/12/2010 8:26      3.39      17850
## 5    536365    84029E        6 1/12/2010 8:26      3.39      17850
## 6    536365     22752        2 1/12/2010 8:26      7.65      17850
##          Country Revenue
## 1 United Kingdom   15.30
```

```
## 2 United Kingdom   20.34
## 3 United Kingdom   22.00
## 4 United Kingdom   20.34
## 5 United Kingdom   20.34
## 6 United Kingdom   15.30
```

```r
# checking if any negative value still exist even after cleaning

Negative_value_check <-sum(orders_clean$Revenue <0)
Negative_value_check
```

```
## [1] 2
```

Interpretation

- here, it showing there are still 2 rows which have negative value, keeping them will lead to misleading revenue analysis. Therefore, we will remove them

Code

```r
# Removing Negative values

orders_clean <- orders_clean[orders_clean$Revenue >0,]

# checking again

sum(orders_clean$Revenue <0)
```

```
## [1] 0
```

Interpretation

- This shows that negative values are successfully removed.This ensures that revenue analysis reflects only the positive transactions.

1.3. Using Inner Join for getting only StockCode Values

```r
## using inner join function

library(dplyr)

orders_final <- orders_clean %>%
  inner_join(product, by = "StockCode")
```

```r
# Using dim() to cross check

dim(orders_clean)
```

```
## [1] 530104      8
```

```r
dim(orders_final)
```

```
## [1] 530104      9
```

Interpretation**

- The above results shows that we haven't lost any rows during inner join, that means product information has been successfully joined.

Code

```r
# Total Revenue

product_revenue <- aggregate(Revenue~StockCode + Description,
                             data = orders_final,
                             sum)
```

```r
# Sorting form highest orders

product_revenue <- product_revenue[order(-product_revenue$Revenue),]
```

```r
# Top 10 product

Top10 <- head(product_revenue,10)
Top10
```

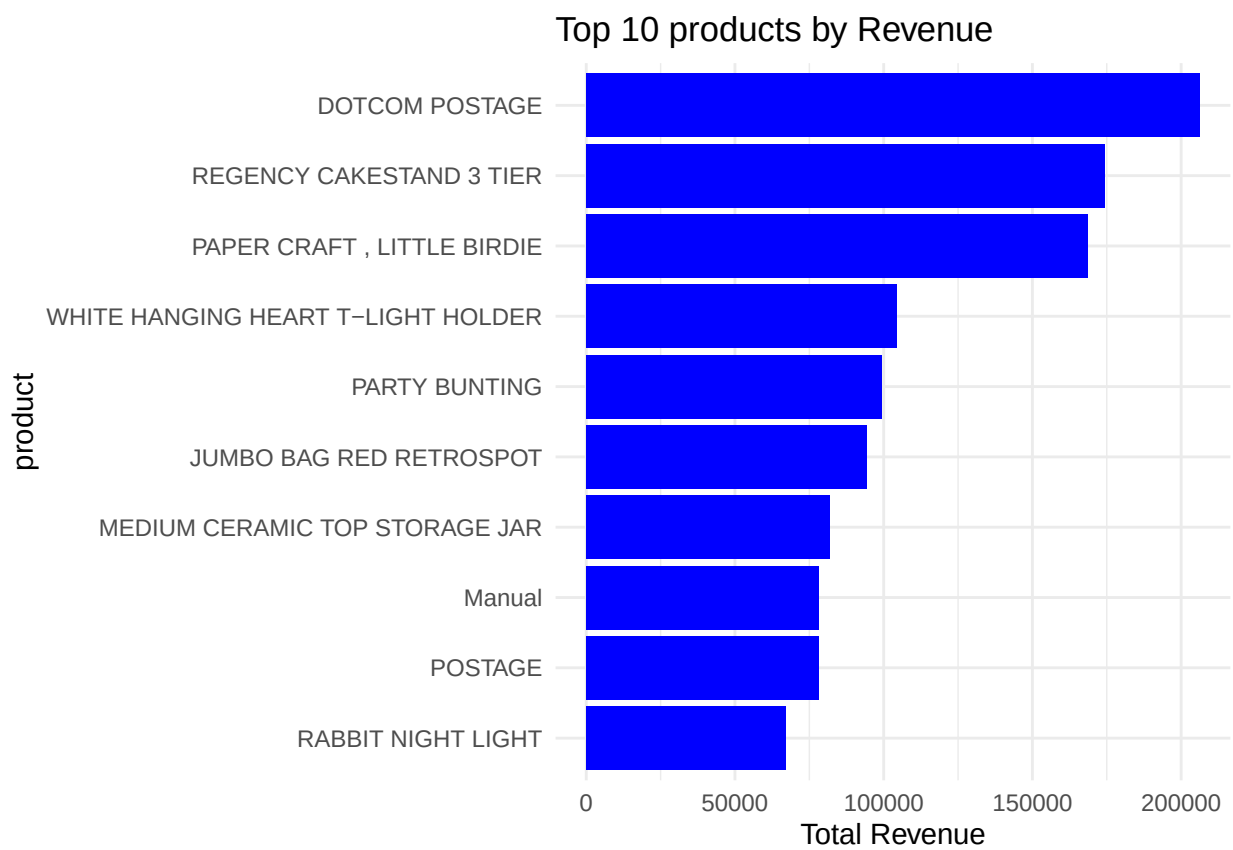
```
##      StockCode                        Description   Revenue
## 1102       DOT                       DOTCOM POSTAGE 206248.77
## 2808     22423             REGENCY CAKESTAND 3 TIER 174484.74
## 2350     23843          PAPER CRAFT , LITTLE BIRDIE 168469.60
## 3750    85123A WHITE HANGING HEART T-LIGHT HOLDER 104518.80
## 2370     47566                       PARTY BUNTING  99504.33
## 1812    85099B              JUMBO BAG RED RETROSPOT  94340.05
## 2035     23166      MEDIUM CERAMIC TOP STORAGE JAR  81700.92
## 2027         M                               Manual  78110.27
## 2646      POST                             POSTAGE  78101.88
## 2695     23084                  RABBIT NIGHT LIGHT  66964.99
```

Interpretation

- here, first we evaluate the product performance by aggregating total revenue at the product level. After that, we sort them in decreasing order and then identify the top 10 products.

Code

```r
# Visualization

library(ggplot2)

ggplot(Top10, aes(x = reorder(Description,Revenue) , y= Revenue))+ geom_bar(stat="identity", fill = "blu
              coord_flip()+
  labs(title = "Top 10 products by Revenue",
       x = "product",
       y = "Total Revenue")+
           theme_minimal()
```



Interpretation

- for Visualization, library(ggplot2) has been used. In code reorder() function is used to control the visualization, so that they appear in a particular order.
- The bar chart shows the top 10 revenue generating after data cleaning. From the above bar graph we can see that, DOTCOM POSTAGE is generating the highest revenue followed by REGENCY CAKESTAND 3 TIER, PAPER CRAFT, LITTLE BIRDIE, etc., are coming out as strongest revenue generating items.
- We see a noticeable drop after 3 products, showing that revenue generation is only limited to few items which include decorative, gift related, home decor and part items.

1.4. Contribution of Top 10 products

```r
# Total Revenues

total_revenue <- sum(orders_final$Revenue)
total_revenue
```

```
## [1] 10666685
```

```r
# Contribution of top 10 products

top10_revenue <- sum(Top10$Revenue)
top10_revenue
```

```
## [1] 1152444
```

```r
# Percentage Contribution

percentage_contribution <- (top10_revenue / total_revenue)*100

percentage_contribution
```

```
## [1] 10.80415
```

Interpretation

- The total revenue is 10666685 while top 10 revenue is 1152444, contributing 10.8% of total revenue.
- This shows that revenue is relatively diversified among many products, instead of concentrating to few number of products.

1.5. Bottom 10 products by Revenue

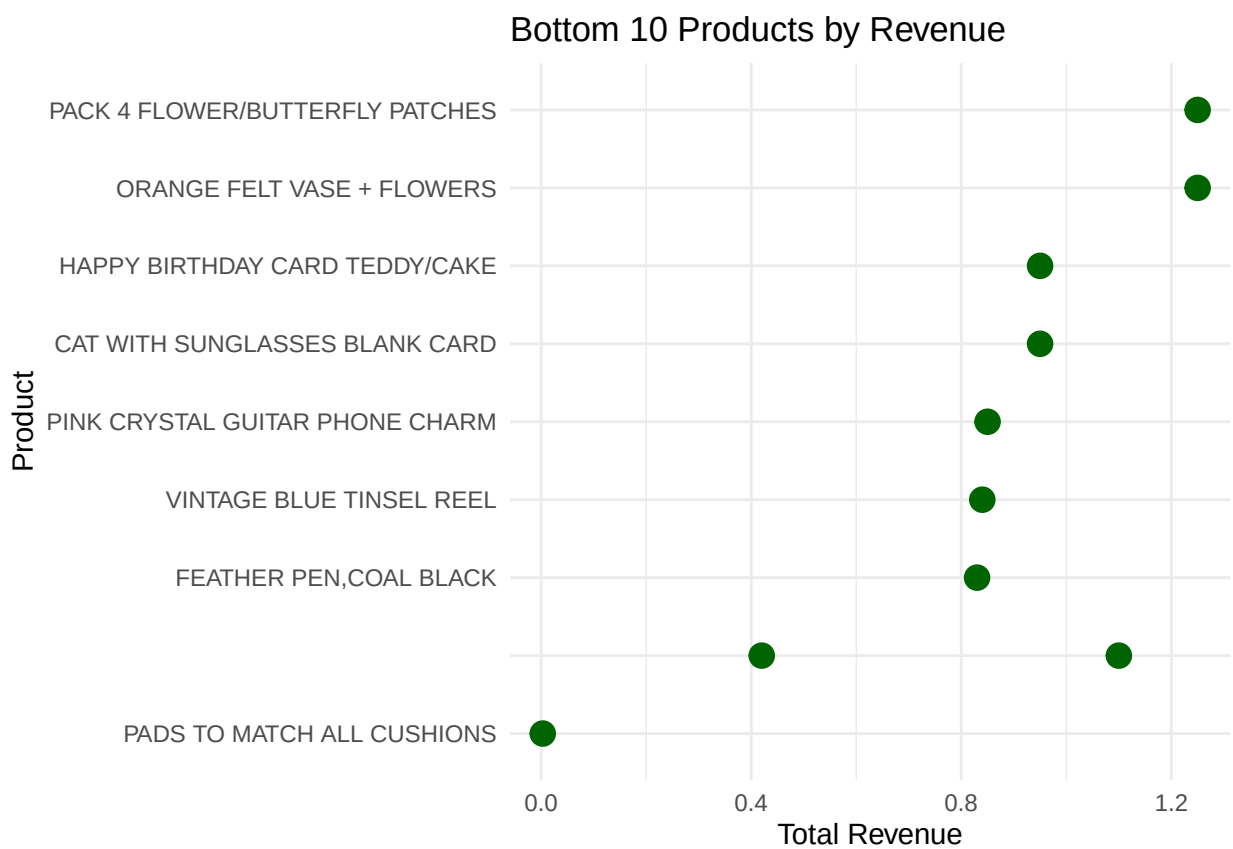
```r
# Bottom 10 Product

bottom10 <- tail(product_revenue,10)
bottom10
```

```
##      StockCode                     Description Revenue
## 2232    84743C      ORANGE FELT VASE + FLOWERS   1.250
## 2263    84569C PACK 4 FLOWER/BUTTERFLY PATCHES   1.250
## 60       84990                                   1.100
## 713     84206B  CAT WITH SUNGLASSES BLANK CARD   0.950
## 1649    84201C  HAPPY BIRTHDAY CARD TEDDY/CAKE   0.950
## 2471     90084 PINK CRYSTAL GUITAR PHONE CHARM   0.850
## 3629     21268        VINTAGE BLUE TINSEL REEL   0.840
## 1251    51014c          FEATHER PEN,COAL BLACK   0.830
## 44       84227                                   0.420
## 2308      PADS      PADS TO MATCH ALL CUSHIONS   0.003
```

```r
# Bottom 10 revenue

bottom10_revenue <- sum(bottom10$Revenue)
bottom10_revenue
```

```
## [1] 8.443
```

```r
# Visualization**

ggplot(bottom10,
       aes(x = Revenue, y = reorder(Description, Revenue))) +
  geom_point(size = 4, colour = "darkgreen") +
  labs(title = "Bottom 10 Products by Revenue",
       x = "Total Revenue",
       y = "Product") +
  theme_minimal()
```



Interpretation

- The bottom 10 contribute negligible to total revenue and contribute almost nothing to overall business revenue. The company can either consider discontinue or promote these products.

1.6. Top 3 products

```r
# Top 3 products

top3 <- head(product_revenue,3)
top3
```

```
##      StockCode                 Description  Revenue
## 1102       DOT              DOTCOM POSTAGE 206248.8
## 2808     22423    REGENCY CAKESTAND 3 TIER 174484.7
## 2350     23843 PAPER CRAFT , LITTLE BIRDIE 168469.6
```

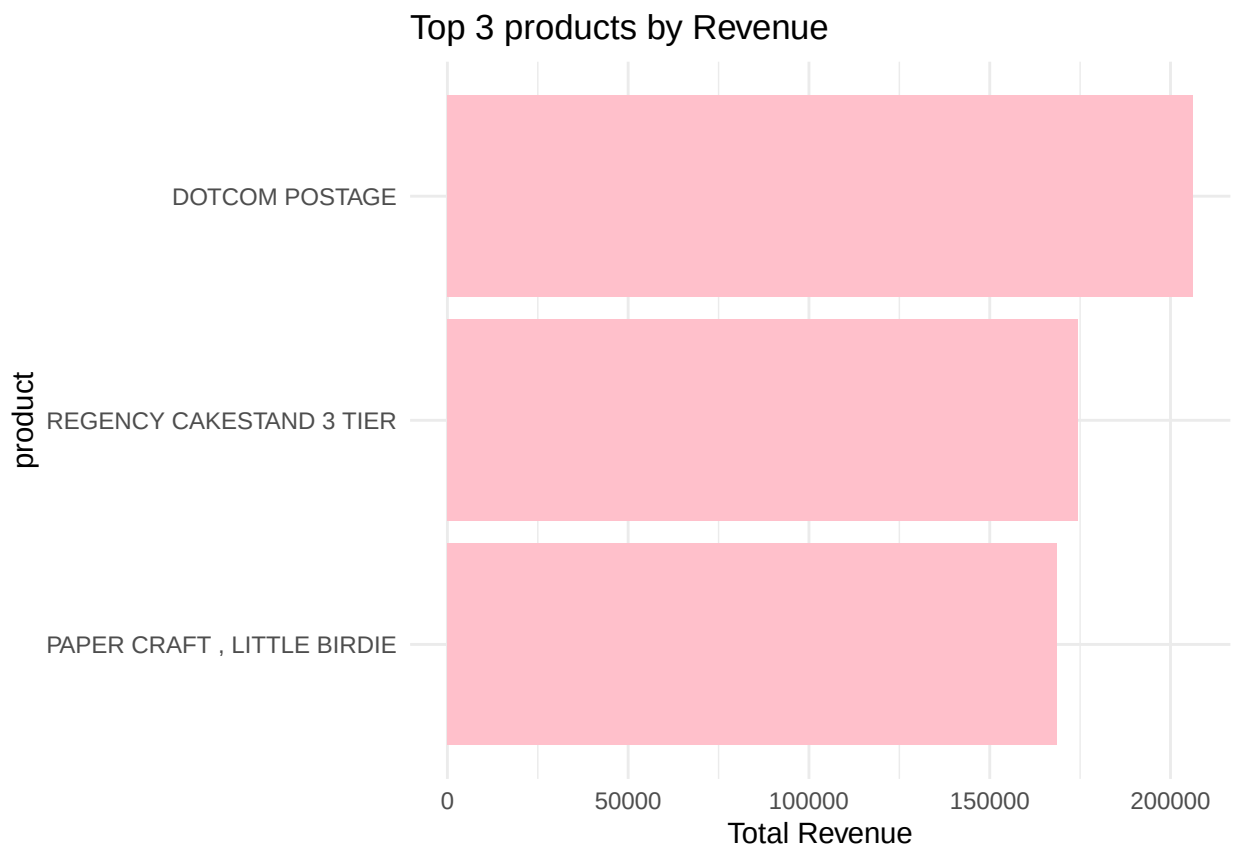
```r
# Revenue of Top 3

top3_revenue <- sum(top3$Revenue)
top3_revenue
```

```
## [1] 549203.1
```

```r
# Percentage Contribution

top3_percentage <- (top3_revenue /total_revenue)*100
top3_percentage
```

```
## [1] 5.148771
```

```r
# Visualization

library(ggplot2)

ggplot(top3, aes(x = reorder(Description,Revenue) , y= Revenue))+ geom_bar(stat="identity", fill = "pinl
             coord_flip()+
  labs(title = "Top 3 products by Revenue",
       x = "product",
       y = "Total Revenue")+
           theme_minimal()
```



Interpretataion

- The top 3 products generate approximately 549203.1, representing 5.15% of total revenue. This indicates that revenue is not heavily concentrated among small number of products, indicating a diversified revenue structure.
- This diversification reduces business risk, as they are not heavily dependent on single product. in addition, top performers include service related revenues to overall sales, showing the contribution by DOTCOM Postage.

Question 2: High Value Customer Analysis

2.1. Each Customer total spending

```
# total spending of each customer

customer_spending <- aggregate(Revenue ~ CustomerID,
                               data = orders_final,
                               sum)
```

```
# Sorting

customer_spending <- customer_spending[order(-customer_spending$Revenue),]
```

```r
#Viewing Top Customers

head(customer_spending)
```

```
##      CustomerID  Revenue
## 1690      14646 280206.0
## 4202      18102 259657.3
## 3729      17450 194550.8
## 3009      16446 168472.5
## 1880      14911 143825.1
## 56        12415 124914.5
```

Interpretation

- Customer ID : 14646 comes out as the highest spending customer. The Revenue contributed by the this customer is 280206. The following highest spending customers are; Customer ID : 18102 , Customer ID : 17450 have revenue contribution, 259657.3 and 194550.8 respectively.
- This shows, small group of customers are contributing significantly to total revenue. These customers should be given priority while deciding important policy like loyalty programs.
- The revenue gap between the top customer and other customers shows the spending pattern behavior of them, it also highlight importance of customer segment in business decision making.

2.2. Top 5 Customers

Code

```r
# Top 5 Highest spending Customers

top5_Customers <- head(customer_spending, 5)
top5_Customers
```

```
##      CustomerID  Revenue
## 1690      14646 280206.0
## 4202      18102 259657.3
## 3729      17450 194550.8
## 3009      16446 168472.5
## 1880      14911 143825.1
```

```r
# Orders of Top 5 Customers

top5_orders <- orders_final[orders_final$CustomerID %in% top5_Customers$CustomerID, ]

dim(top5_orders)
```

```
## [1] 8522    9
```

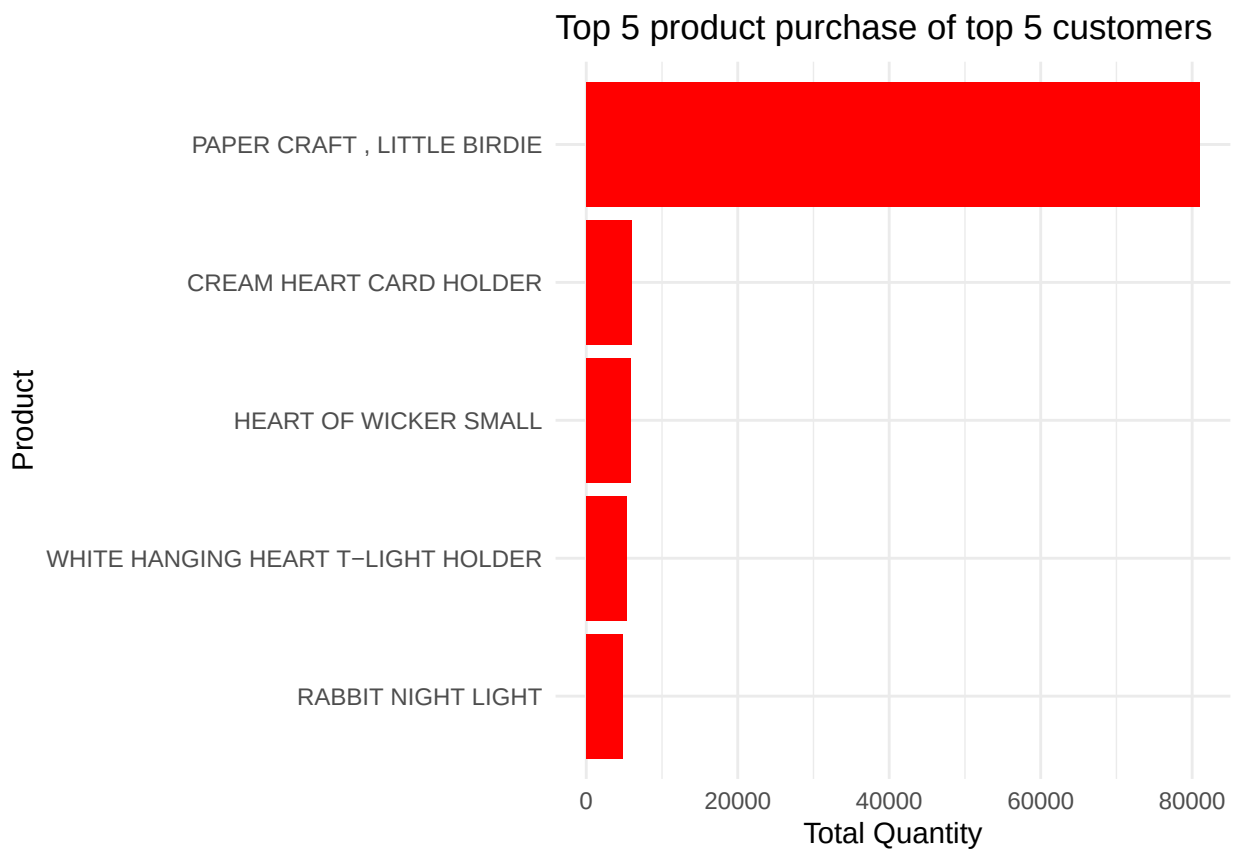
```r
# top 5 Customer's total quantity

top5_quantity <- aggregate(Quantity ~ Description, data = top5_orders, sum)
```

```r
# Sort

top5_quantity <- top5_quantity[order(-top5_quantity$Quantity), ]
```

```r
# top 5  product

top5_quant<-head(top5_quantity, 5)
top5_quant
```

```
##                              Description Quantity
## 1187        PAPER CRAFT , LITTLE BIRDIE    80995
## 436             CREAM HEART CARD HOLDER     5978
## 815               HEART OF WICKER SMALL     5833
## 1909 WHITE HANGING HEART T-LIGHT HOLDER     5300
## 1336                 RABBIT NIGHT LIGHT     4849
```

```r
# Visualization

ggplot(top5_quant , aes(x=reorder(Description, Quantity), y= Quantity))+
  geom_bar(stat = "identity", fill = "red")+
  coord_flip()+
  labs(title = "Top 5 product purchase of top 5 customers",
       x = "Product",
       y = "Total Quantity")+
  theme_minimal()
```



Interpretation

- Top-five highest spending customers shows a strong concentration in a small number of products. "PAPER CRAFT, LITTLE BIRDIE" comes out as dominating purchase with 80995 units being purchased, which is way higher than any other product.
- Other products "CREAM HEART CARD HOLDER", "HEART OF WICKER SMALL", "WHITE HANGING HEART LIGHT HOLDER" presents a lower demand. Its shows clear preference of customers, highlighting that customers are not buying randomly.
- The high demand of "PAPER CRAFT , LITTLE BIRDIE" shows bulk purchasing behavior of customers towards this product. This pattern helps company to make customer segment strategies, offer discount and implement loyalty program.

2.3. Business Implication

- From the above analysis, we can say that small group of customers contribute more in total revenue. Customer id 14646 is generating 280000 revenue alone. It highlight the importance of retaining top customers.
- Company should introduce programs that targeted to these customers, do personalized promotions and pricing strategies for these customers.
- From the analyse, we also see that certain products have dominance and has high demand along with that they are also high risk products as any disruption in their supply can lead to high losses for company. Due to this they have to monitor the supply chain regularly.

Question 3: Global Market Analysis

3.1. Total Revenue country wise

```
# Total Revenue by country

Country_revenue <- aggregate(Revenue~ Country, data =orders_final, sum)
```

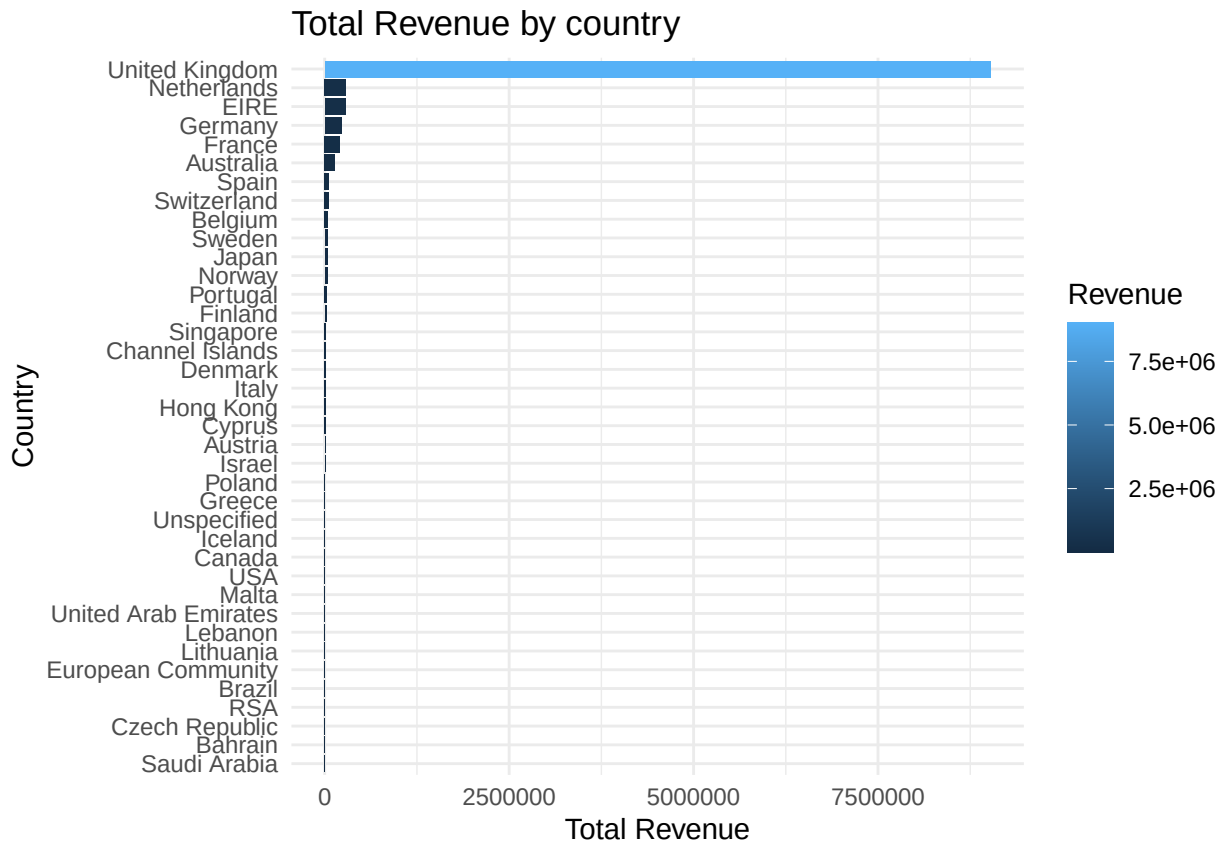
```
# sort

Country_revenue <- Country_revenue[order(-Country_revenue$Revenue),]
```

```
# Details

head(Country_revenue)
```

```
##           Country   Revenue
## 36 United Kingdom 9025222.1
## 25    Netherlands  285446.3
## 11           EIRE  283454.0
## 15        Germany  228867.1
## 14         France  209715.1
## 1       Australia  138521.3
```

3.2. Comparing Revenue among Countries

```
# Visualization

ggplot(Country_revenue,
       aes(x=reorder(Country, Revenue),
           y = Revenue, fill = Revenue))+
  geom_bar(stat="identity")+
  coord_flip()+
  labs(title = "Total Revenue by country",
       x = "Country",
       y = "Total Revenue")+
  theme_minimal()
```



Interpretation

- From the above graph, it is clear that United Kingdom has generated highest revenue (i.e. 9025222.1) as compared to other countries and the difference between their revenue is very large. This shows that business is highly concentrated in UK.
- Uk is followed by Netherlands, Germany, France, Australia, etc.,having revenue 285446.3, 283454.0, 228867.1, 209715.1, 138521.3 respectively.
- The huge gap between Uk and other countries shows that it has strong domestic customers base while limited international exposure.
- The company need to adopt a strategy where they can diversify the revenue generation sources and reduces risk. They need to expand their international market limits and make a string presence in potential market of growth like Germany, France, etc., In the long term it will generate more revenue and growth of the company and reduces geographic risk.

3.3. Highest Revenue generated product in each country

```
# Quantity by product and country

country_product_quantity <- aggregate(Quantity ~Country+ Description,
                                      data = orders_final,sum)
```

```r
# Sort

country_product_quantity<- country_product_quantity[order(country_product_quantity$Country,
                                                          -country_product_quantity$Quantity),]
```

```r
# Top Product of each country

Top_product_Country <- country_product_quantity[!duplicated(country_product_quantity$Country),]
```

```r
# Viewing the Top Countries

head(Top_product_Country,10)
```

```
##               Country                        Description Quantity
## 9795        Australia            MINI PAINT SET VINTAGE      2952
## 14599         Austria   SET 12 KIDS COLOUR  CHALK STICKS      288
## 7874          Bahrain         ICE CREAM SUNDAE LIP GLOSS       96
## 10871         Belgium    PACK OF 72 RETROSPOT CAKE CASES      480
## 4694           Brazil               DOLLY GIRL LUNCH BOX       24
## 13715          Canada         RETRO COFFEE MUGS ASSORTED      504
## 12663 Channel Islands                       RAIN PONCHO       407
## 7443           Cyprus     HEART DECORATION PAINTED ZINC       384
## 18842  Czech Republic WOODEN STAR CHRISTMAS SCANDINAVIAN       72
## 12841         Denmark             RED  HARMONICA IN BOX       288
```

Interpretation

- The above table shows that different countries have different preferred product. Hence, there is no universal best -selling product. This also explains customer behavior which vary differently according to different region.
- Considering that, the company should focus on different product according to different region depending on the local demand, taste and buying pattern.

Question 4: Temporal Analysis and Trends

4.1. Extracting Month and Year from the Invoice Date

```r
# Checking the format of the Date

head(orders_clean$InvoiceDate, 5)
```

```
## [1] "1/12/2010 8:26" "1/12/2010 8:26" "1/12/2010 8:26" "1/12/2010 8:26"
## [5] "1/12/2010 8:26"
```

```r
# Using the correct data set for extracting Invoice date

orders_final<- orders_clean
```

```r
# Convert Invoice Date correctly
orders_final$InvoiceDate <- as.POSIXct(
  orders_final$InvoiceDate,
  format = "%d/%m/%Y %H:%M",
  tz = "UTC"
)
```

```r
# Checking if Dates are converted in right format or not?

head(orders_final$InvoiceDate, 5)
```

```
## [1] "2010-12-01 08:26:00 UTC" "2010-12-01 08:26:00 UTC"
## [3] "2010-12-01 08:26:00 UTC" "2010-12-01 08:26:00 UTC"
## [5] "2010-12-01 08:26:00 UTC"
```

```r
sum(is.na(orders_final$InvoiceDate))   # should be 0
```

```
## [1] 0
```

```r
# Extracting Year and Month

orders_final$Year  <- format(orders_final$InvoiceDate, "%Y")
orders_final$Month <- format(orders_final$InvoiceDate, "%m")
table(orders_final$Year)
```

```
##
##   2010   2011
##  41480 488624
```

```r
# Monthly Revenue Table

monthly_revenue <- aggregate(Revenue~ Year+Month, data =orders_final, sum)

monthly_revenue <- monthly_revenue[order(monthly_revenue$Year, monthly_revenue$Month),]

head(monthly_revenue,10)
```

```
##    Year Month   Revenue
## 12 2010    12  823746.1
## 1  2011    01  691364.6
## 2  2011    02  523631.9
## 3  2011    03  717639.4
## 4  2011    04  537808.6
## 5  2011    05  770536.0
## 6  2011    06  761739.9
## 7  2011    07  719221.2
## 8  2011    08  759138.4
## 9  2011    09 1058590.2
```

```r
# Highest Month

monthly_revenue[which.max(monthly_revenue$Revenue),]
```

```
##    Year Month Revenue
## 11 2011    11 1509496
```

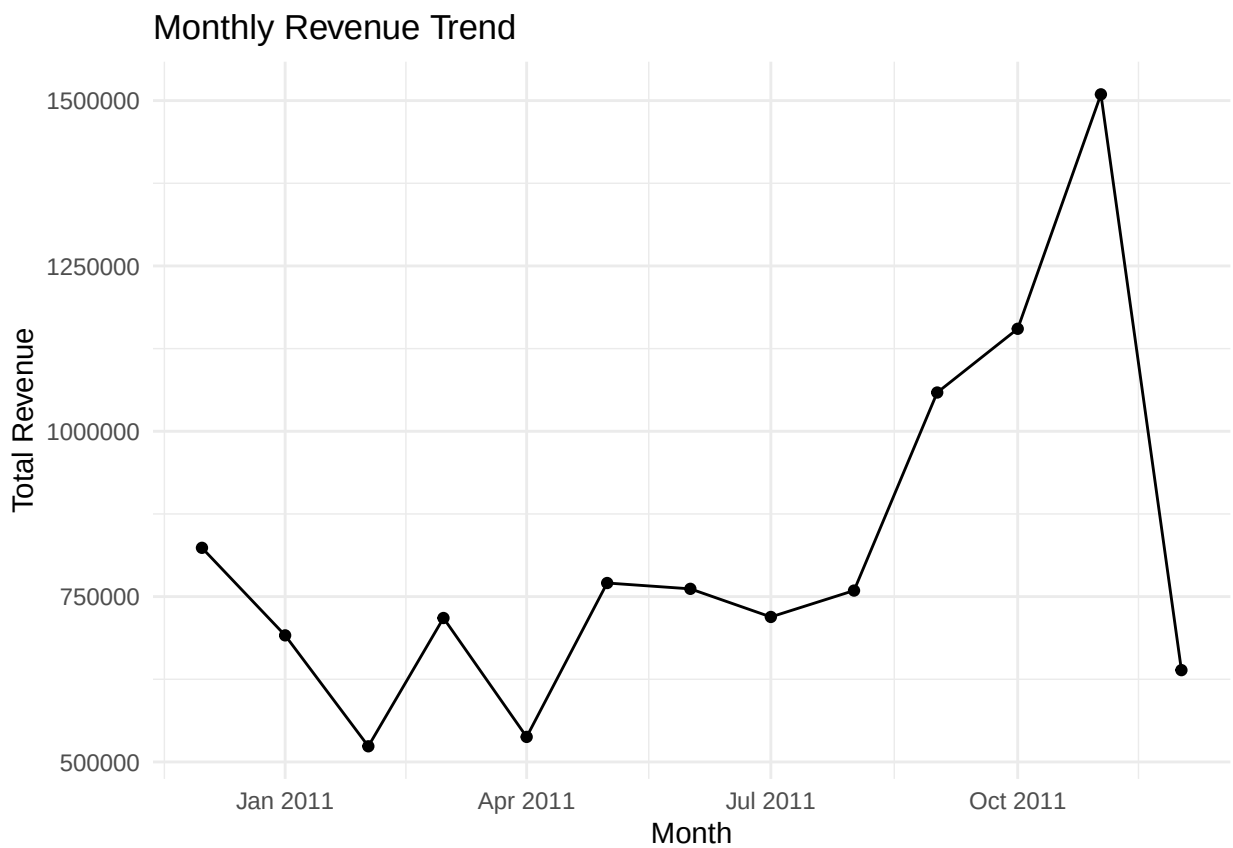
4.2. Month date column

```r
monthly_revenue$MonthDate <- as.Date(paste0(monthly_revenue$Year,"-",
                                            monthly_revenue$Month,"-01"))

head(monthly_revenue[,c("Year","Month","Revenue","MonthDate")])
```

```
##    Year Month  Revenue  MonthDate
## 12 2010    12 823746.1 2010-12-01
## 1  2011    01 691364.6 2011-01-01
## 2  2011    02 523631.9 2011-02-01
## 3  2011    03 717639.4 2011-03-01
## 4  2011    04 537808.6 2011-04-01
## 5  2011    05 770536.0 2011-05-01
```

4.3. Visualization

```r
# Visualization

ggplot(monthly_revenue, aes(x=MonthDate, y= Revenue))+
  geom_line()+
  geom_point()+
  labs(title = "Monthly Revenue Trend",
       x ="Month",
       y= "Total Revenue")+
  theme_minimal()
```



Interpretation

- The above picture shows Monthly Revenue trend, the figure shows that there is a sales decline during the February and March period. Then after April 2011, the sales start soaring again. The sales jump high in November 2011. But then again it drop sharply in December 2011.
- From the above pattern, we can see that there is a seasonality in customer demand. It is high during the end of the year during the holidays and pre-Christmas period.

4.4. Monthly product Revenue

```
# merging orders_final with orders_final2

orders_final2 <- merge(orders_final, product, by= "StockCode")

colnames(orders_final2)
```

```
##  [1] "StockCode"   "InvoiceNo"   "Quantity"    "InvoiceDate" "UnitPrice"
##  [6] "CustomerID"  "Country"     "Revenue"     "Year"        "Month"
## [11] "Description"
```

```r
# Monthly Product Revenue

monthly_product_revenue <- aggregate(Revenue~ Year + Month + Description,
                                     data = orders_final2,
                                     sum)

head(monthly_product_revenue)
```

```
##   Year Month Description Revenue
## 1 2011    01             3891.56
## 2 2011    02              867.53
## 3 2011    03             1378.59
## 4 2011    04             4011.89
## 5 2011    05             1109.23
## 6 2011    06              681.46
```

4.5. Bottom Products

```r
#  Identifying bottom products

library(dplyr)

bottom_products <- monthly_product_revenue %>%
  arrange(Year, Month, Revenue) %>%
  group_by(Year, Month) %>%
  slice_head(n=5)

head(bottom_products,10)
```

```
## # A tibble: 10 x 4
## # Groups:   Year, Month [2]
##    Year  Month Description                          Revenue
##    <chr> <chr> <chr>                                  <dbl>
##  1 2010  12    "ASSORTED MONKEY SUCTION CUP HOOK"      0.42
##  2 2010  12    "METAL BASE FOR CANDLES"                0.42
##  3 2010  12    "SILICON CUBE 25W, BLUE"                0.42
##  4 2010  12    "SILICON STAR BULB  BLUE"               0.42
##  5 2010  12    "HEARTS WRAPPING TAPE "                 0.65
##  6 2011  01    "ANGEL DECORATION STARS ON DRESS"       0.42
##  7 2011  01    "ASSORTED CIRCULAR MOBILE"              0.42
##  8 2011  01    "ASSTD RASTA KEY-CHAINS"                0.42
##  9 2011  01    "THE KING GIFT BAG"                     0.65
## 10 2011  01    "LETTER \"F\" BLING KEY RING"           0.83
```

```r
# Counting Repeating Items

bottom_counts <- as.data.frame(table(bottom_products$Description))

# Sort

bottom_counts <- bottom_counts[order(-bottom_counts$Freq),]
```

```r
head(bottom_counts,10)
```

```
##                                Var1 Freq
## 10 ASSORTED COLOUR SUCTION CUP HOOK    3
## 41       PADS TO MATCH ALL CUSHIONS    3
## 47          SILICON STAR BULB  BLUE    3
## 11 ASSORTED MONKEY SUCTION CUP HOOK    2
## 14    BLUE FLOCK GLASS CANDLEHOLDER    2
## 29            GLOW IN DARK DOLPHINS    2
## 32           HARDMAN MUG 3 ASSORTED    2
## 36        LETTER "F" BLING KEY RING    2
## 40               MINI JIGSAW PURDEY    2
## 45 PINK/GREEN FLOWER DESIGN BIG MUG    2
```

Interpretation

- From the above table, we can see that there are some certain products that are appearing consistently among the lowest selling items.
- Products like "Assorted Color Suction Cup Hook", "Pads To Match All Cushions", "Silicon Star Bulb Blue" has appeared 3 times, and other products like "Assorted Monkey Suction Cup Hook", "Blue Flock Glass Candleholder", etc., has appeared 2 times.
- The repeated appearance of these products in the bottom indicating their weak performance. This shows that they have low demand and has very limited customer approach.
- The company can consider various options for these products like; discontinuation of selling these products so that they can invest that money on popular products. Second, they can spend money on promotion of these products so that people get to know about these products and their sales might increase. Third; they can clear these products on discount, so that they can clear inventory or storage space and utilize that space for other products.

---

5. Conclusion

- The above analysis provide insights related to performance of products, behavior of customers, revenue distribution, sales trends.
- The products revenue analysis shows that revenue generation is mostly limited to few items. Customer level analysis shows that a small group of customers contribute more to total revenue, indicating the need for introducing strategy to target high value customers to retain them to longer period by introducing loyalty programs.
- From the geographical perspective, we see that Uk has dominance there, on the other hand, other neighboring countries are contributing but a very small share. There is a huge potential for growing internationally. Under Temporal analysis, we see the trends of sales, which was high in November and them drop sharply in December.
- In conclusion, we can say that business shows strong seasonal behavior, identifying high value customers, sales that are limited to particular area and some popular products. Based on that we can say that the company need to introduce strategy which focused on customer retention, product promotion, seasonal planning and spreading internationally.

---

6. Git WorkFlow**

- During the project, two branches are created, The following are the branches name:

1. fix-Q1-1.3.-interpretation
2. fix-bottom-10-visualization

- In “fix-Q1-1.3.-interpretation”, I encountered a small mistake that is, I mistakenly add “/” at the end of interpretation. To rectify it, I made a branch and corrected this by removing it from there.
- In “fix-bottom-10-visualization”, at first I visualize bottom 10 by bar graph but then I thought of using dot plot, to create better and more correct picture.I made a branch and corrected this by adding new type of visualization.
- All changes are merge back to main after validation.

I confirm that: - I understand all code submitted. - I can explain my solution if required.